The development of turn-taking in Kata Kolok through social interaction

Hannah Lutzenberger [*1,2]

[*]Corresponding Author: hannah.lutzenberger@hu-berlin.de
[1]Department of Linguistics, Stockholm University, Stockholm, Sweden
[2]Department of Sign Languages, Humboldt University Berlin, Berlin, Germany

Turn-taking is one of the fundamental skills children need to learn to become competent conversationalists. Across different spoken and signed languages, adults take turns effortlessly, monitoring their interlocutors' turn while planning and carefully timing their own response to achieve minimal gaps and overlaps between turns (Stivers et al. 2009, De Vos et al. 2015). Children's structure and timing of turns gradually develop alongside their linguistic and pragmatic skills (Casillas et al. 2016). Before age 6, response latencies are common; while adult interlocutors may manage slow responses, these delays may cause non-coordinated turns and misunderstandings when there are more than two conversation partners or in peer-to-peer conversations (Ervin-Tripp, 1979). Following the ideas of the Interaction Engine Hypothesis (Levinson 2006), this skills I fundamental is developed and practiced through social interaction, i.e. face-to-face conversations.

I investigate the development of turn-taking in face-to-face interaction in Kata Kolok, the indigenous sign language of a single village in Bali, Indonesia, where multi-party conversations are the norm. Two key aspects of this signing community make it unusual. First, sustained high incidences of deafness have led to six generations of deaf and hearing signers, which provides children access to a **diversity** of signing interlocutors (Marsaja 2008). Second, communal child-rearing practices mean that most conversations available to children involve **multiple** signing interlocutors. As such, it allows us to study the development of turn-taking from two perspectives: 1) the glossogenetic process within Kata Kolok over several generations of signers, and 2) the ontogenetic process of turn-taking in deaf children.

I analyse longitudinal corpus data from adult signers of Kata Kolok and four deaf children from the Kata Kolok Child Signing Corpus, a collection of naturalistic recordings of children who acquire Kata Kolok from their deaf caregivers. Using conversation analysis, I compare structure and timing of 200 question-answer sequences from conversations among adults and 1,200 question-answer sequences with a child interlocutor (i.e., 300 questions per children distributed across 10 longitudinal recordings between 1-3 years old). For structure, I focus on question/response type, turn complexity and interactional context and for timing, I focus on gaps and overlap.

Preliminary findings suggest six major observations:

1) Overlap in questions and answers among adult interlocutors is high, possibly facilitated by routinized question-answer structure shaped by shared context and culture.
2) Polar questions prevail among both adult and child interlocutors. Yet, complexity of questions and responses gradually increases as children get older, proceeding from short to longer, information-seeking questions and responses.
3) Young children are often prompted with short polar questions that are embedded in familiar routines, such as EAT FINISH ('have you eaten'). These are repeated (over time and within the same interactive sequence) and deeply culturally entrenched, allowing children to hone the structure and timing of turns.
4) Possibly due to challenges of coordinating and timing visual attention, young children only gradually partake in multiparty settings, and if so, the conversations often involve peers and/or few adult interlocutors. Indeed, within medium-sized to large multiparty settings, children often are passive participants and/or initiate dyadic conversations.
5) The timing of question-answer sequences is influenced by conversational setting and age: conversations with younger children are generally slower (i.e. longer gaps between turns) with less overlap than when adult interlocutors are involved.
6) Often, even among adult interlocutors, questions are not truly seeking new information and instead elicit confirmation of already known information. This may well be influenced by cultural norms and the high degree of shared context. Given the human desire of social interaction, face-to-face conversations center around daily experiences, even though these experiences are often already shared.

This study presents critical insights into the development of turn-taking: 1) effects of culture and shared context on turn-taking, 2) development of turn-taking in children, and 3) the relation between language-level characteristics of turn-taking and developmental patterns of children. This unprecedented study into turn-taking in a small, rural sign language from the Global South, using a unified method in adults and children adds significant depth to our understanding of social interaction.

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